

BUSINESS ANALYSIS REPORT

Airbnb NYC Listings

End-to-end analysis of **102,036 cleaned listings** data cleaning (Python: NumPy, Pandas), SQL, Power BI

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Project at a Glance

102K

Listings Analyzed

Cleaned dataset across all 5 NYC boroughs

10

Business Questions

Investigated via SQL

22

Final Columns

Down from 26 after cleaning

541

Duplicates Removed

Plus invalid rows dropped

Cleaning Pipeline



Identify Issues



Correct Types



Handle Outliers



Finalize Dataset

Every decision prioritized honest, defensible outcomes over technical convenience.

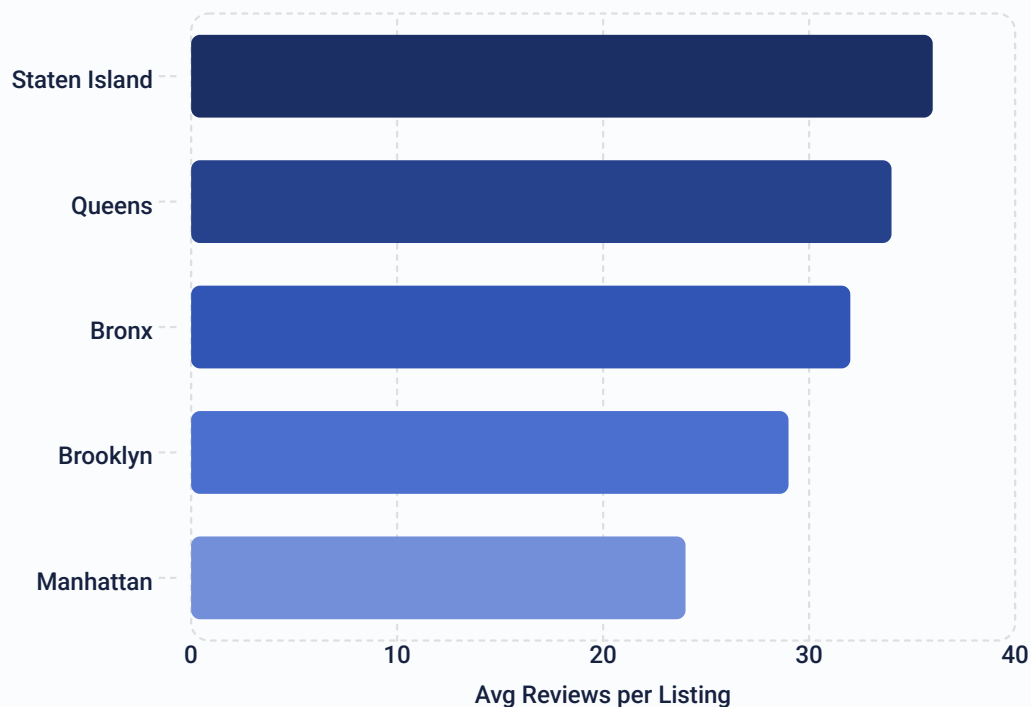
10 Questions Driving the Analysis

- Which areas have the highest review activity?
- How does availability vary by neighborhood?
- Which hosts have the most listings?
- Which neighborhoods offer the best price-to-review balance?
- What percentage of listings are instant-bookable?
- Which neighborhood groups have the highest average prices?
- Which room types generate the most listings?
- How does cancellation policy relate to pricing?
- What factors appear associated with higher prices?
- Which areas have the highest average availability?

KEY FINDING

Engagement Is Inversely Tied to Listing Volume

Borough



What This Means

Staten Island and Queens show the highest reviews per listing despite far fewer total listings than Manhattan.

However, these also have the **highest availability**, indicating **historical engagement**, not current demand.

KEY FINDING

Reviews and Availability Move Together

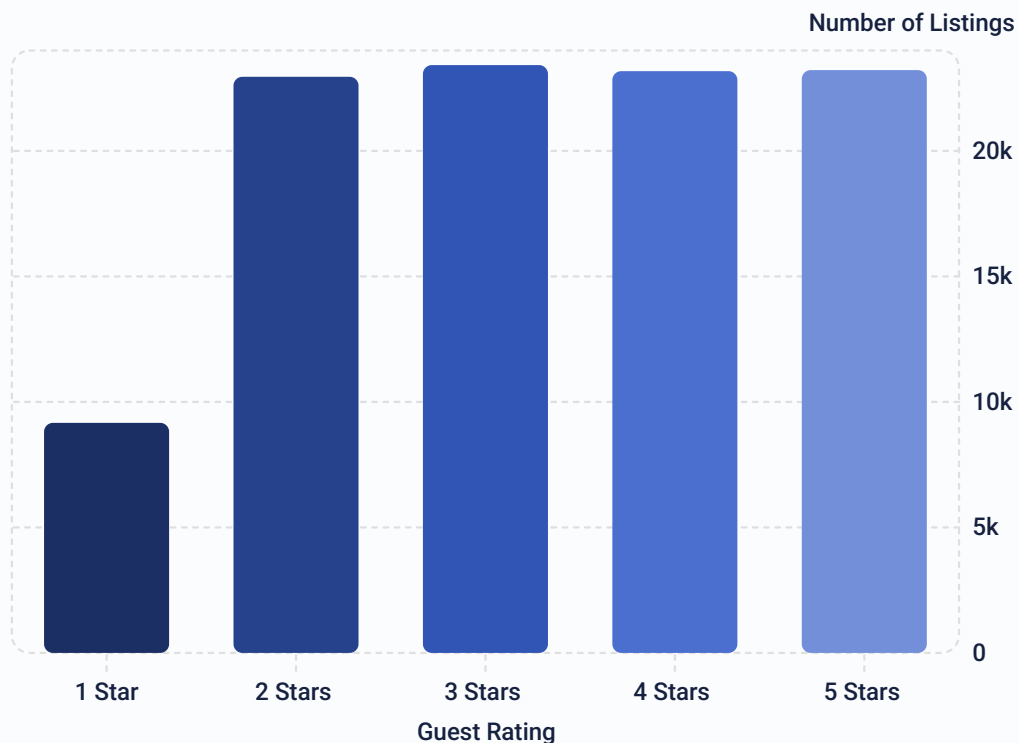
■ Avg Reviews ■ Avg Availability (Days)



Interpretation

Both metrics move together across boroughs, revising the Staten Island story from **"high demand"** to **"established but currently quieter listings."**

Price Data Shows No Real-World Signal



Red Flags

- IQR test: zero outliers, suspiciously uniform percentiles
- No relationship with room type, borough, or cancellation policy
- 1-star listings priced higher than 5-star
- Dozens of listings share exact max price: \$1,200



Price column likely synthetically generated. No recommendations rely on it.

Geographic & Neighborhood Insights



Listing Density Map

Clear clustering around Manhattan and Brooklyn, visible at a glance, not from a table.

Sample Size Correction

Filtering to 20+ listings per neighborhood removed distortion from areas with only 1-2 listings.

RECOMMENDATIONS

Three Actions for Stakeholders

1

Investigate Staten Island & Queens

High reviews + high availability = established listings, not current demand. Review for staleness; assess marketing opportunity.

2

Audit Host Listings Count

Column contradicts verifiable host_id counts. Do not use until source is confirmed.

3

Replace the Price Column

No realistic relationship with any variable. Future pricing work requires a verified source.

CONCLUSION

Key Takeaways

What Was Found

- Lower-volume boroughs show higher reviews per listing
- High reviews reflect historical engagement, not current demand
- Price column and host count column are unreliable

What Was Demonstrated

- End-to-end workflow: Pandas → SQL → Power BI
- Testing assumptions over accepting data at face value
- Documenting limitations as a core part of analysis